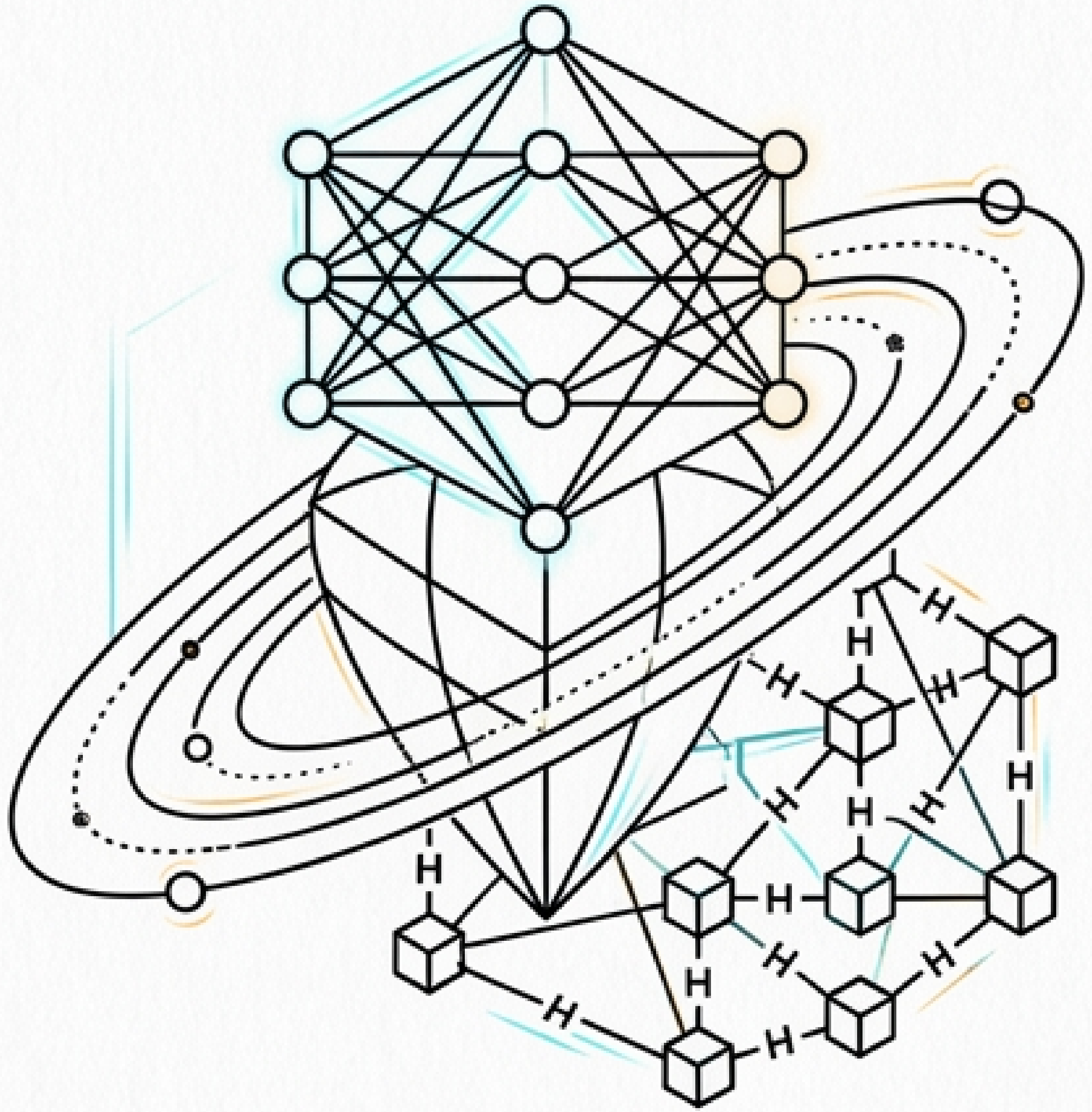
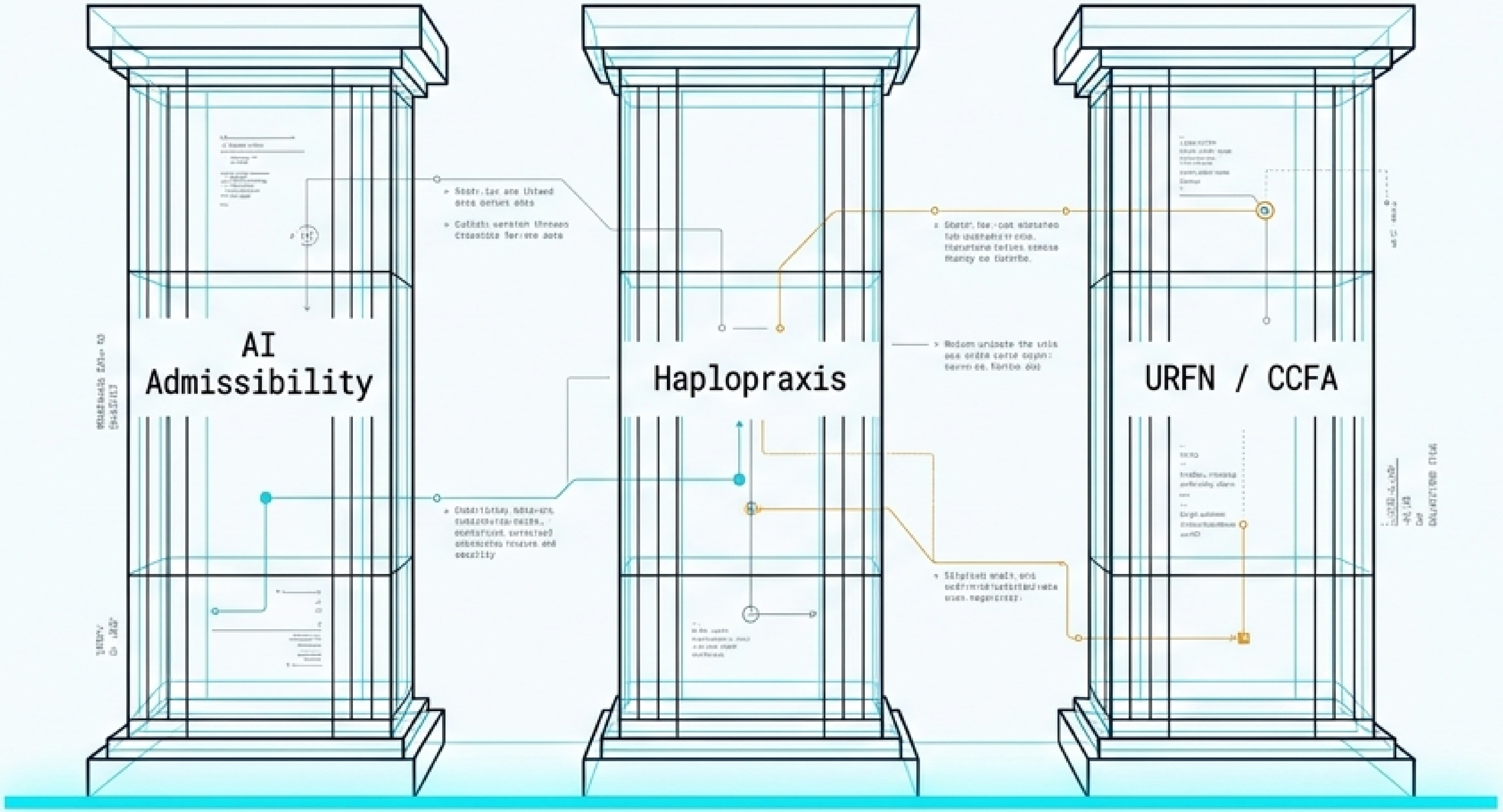


Architectures of History & State

THREE MODELS OF RELATIONAL SYSTEMS

A synthesis of LLM interpretability, decentralized constitutional design, and emergent ontology.

State is an Illusion.
History and Routing are Primary.



BLOCK 01 // MANIFESTO

Static storage is a flawed metaphor across computational domains.

> OBSOLETE: STATE AS PERSISTENCE

BLOCK 02 // ONTOLOGY

Whether fine-tuning an LLM, designing a distributed world, or building a universal resource fabric, the underlying truth holds: the event log is the ontology. State is merely a derived view.

> EVENT LOG: IMMUTABLE TRUTH

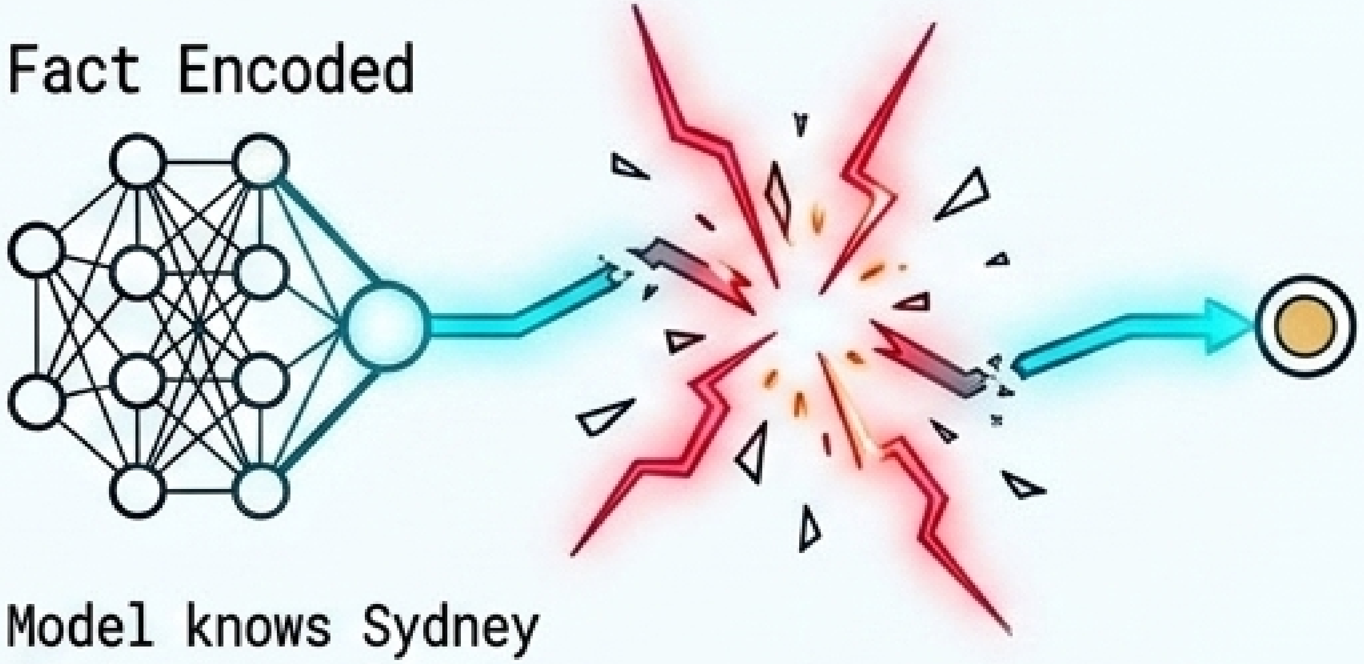
BLOCK 03 // CURATION

This deck curates three architectural models that solve for continuity, trajectory, and exact versus inexact representation.

> MODELS: CONTINUITY, TRAJECTORY, REPRESENTATION

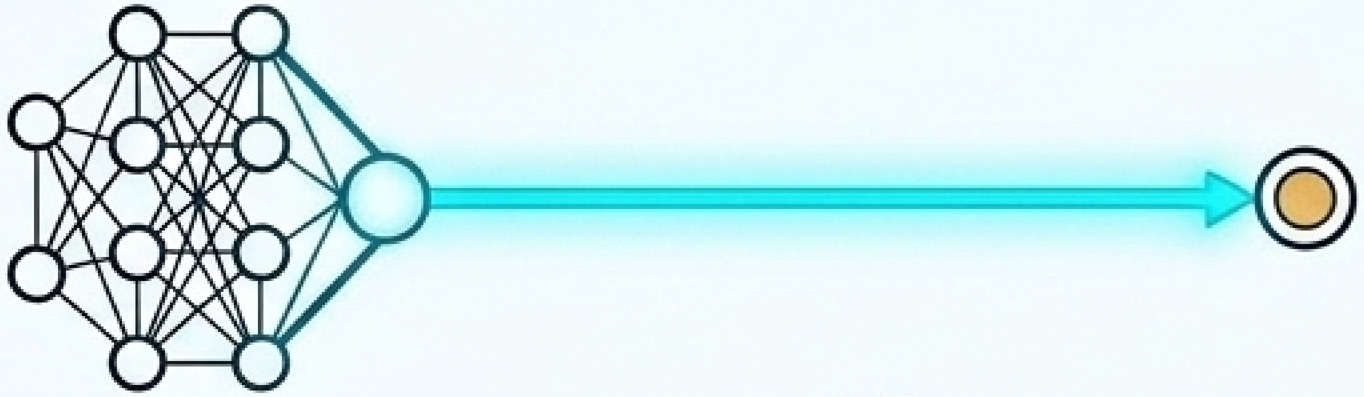
The Knowing-Using Gap

Fact Encoded



Model knows Sydney
is in Australia.

Fact Encoded



Model knows Sydney
is in Australia.

Model cannot answer:
What is the capital
of the country where
Sydney is?

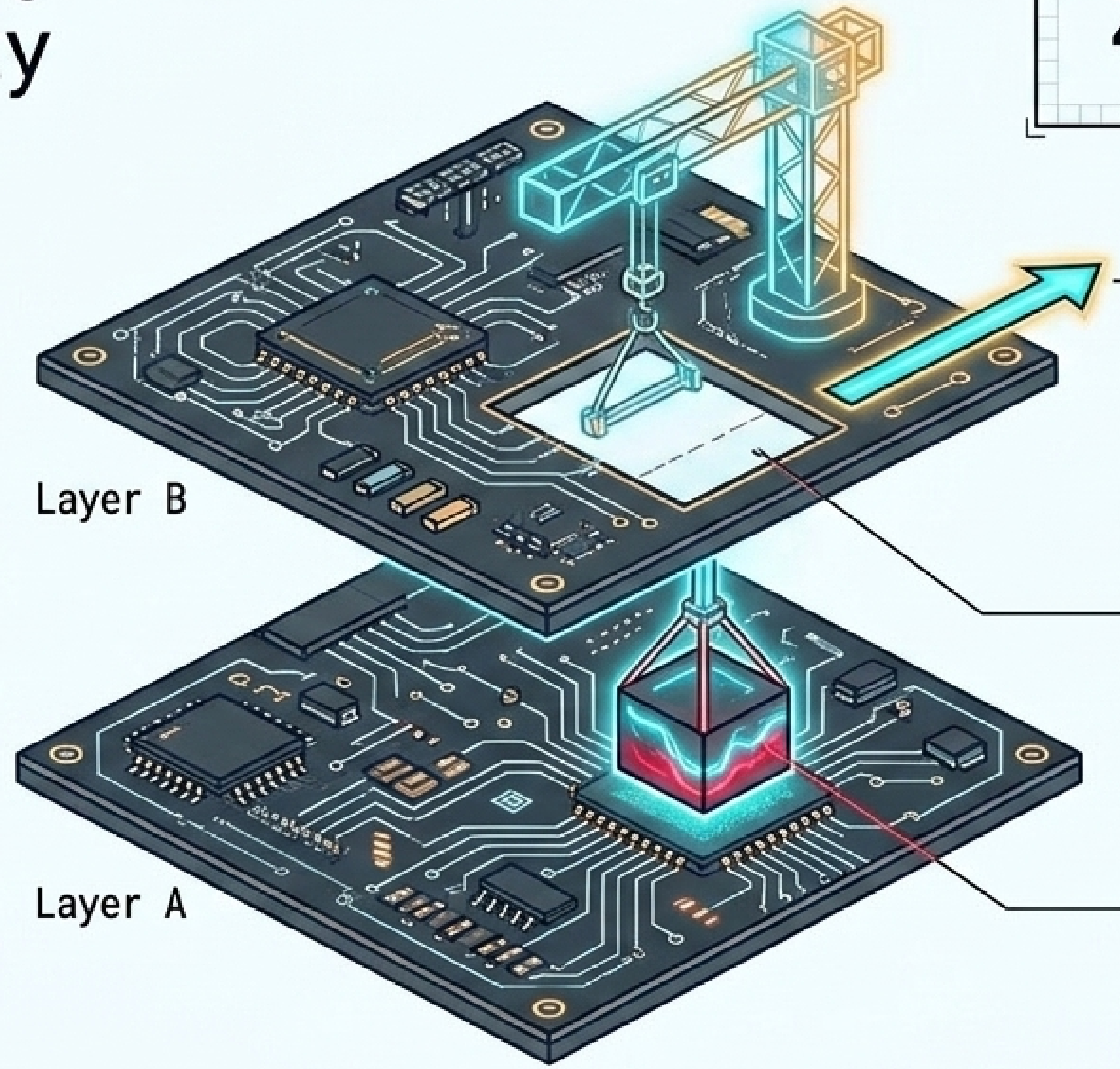
A state's membership in a system's representational space does not entail its membership in the effective domain of the operators that need it. A fact can be memorized perfectly, yet remain unusable for downstream reasoning.

Self-Patching and Directed Admissibility

$$z(i) \rightsquigarrow R_j$$

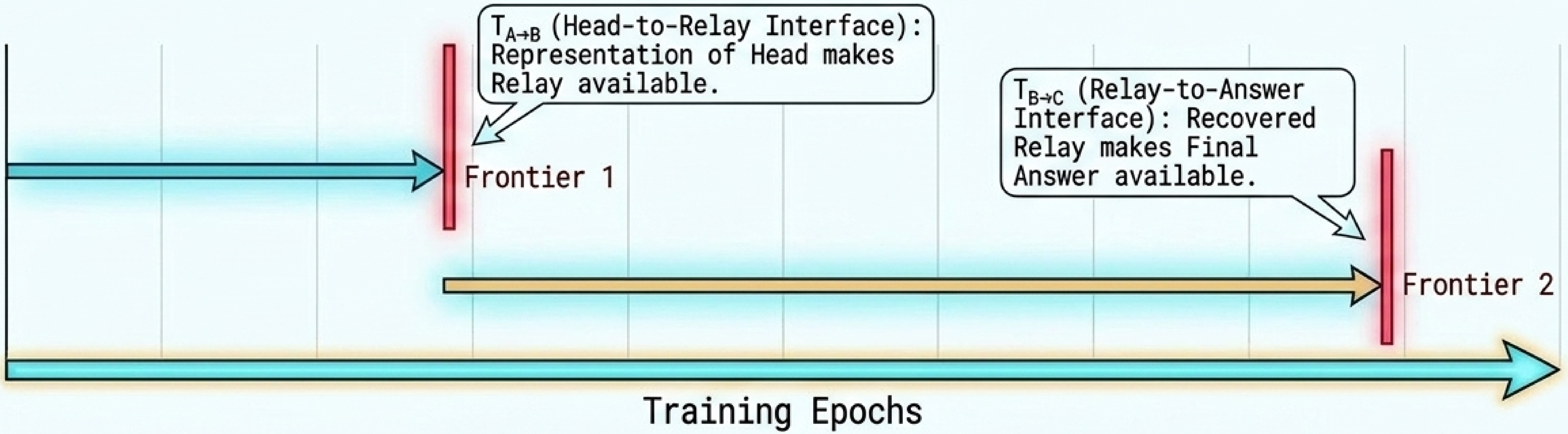
Knowledge isn't a stored object in a room.
It is a relational capacity.

Self-patching proves that the missing ingredient isn't the fact itself, but the active realization of the route that delivers it to the exact layer where the downstream continuation can act on it.



The Decomposition of Chaining

$$T_{gen} = \max(T_{A \to B}, T_{B \to C})$$



Chaining tasks do a not suffer one delayed generalization crisis—they suffer two. Generalization only occurs once the final required admissibility relation becomes available.

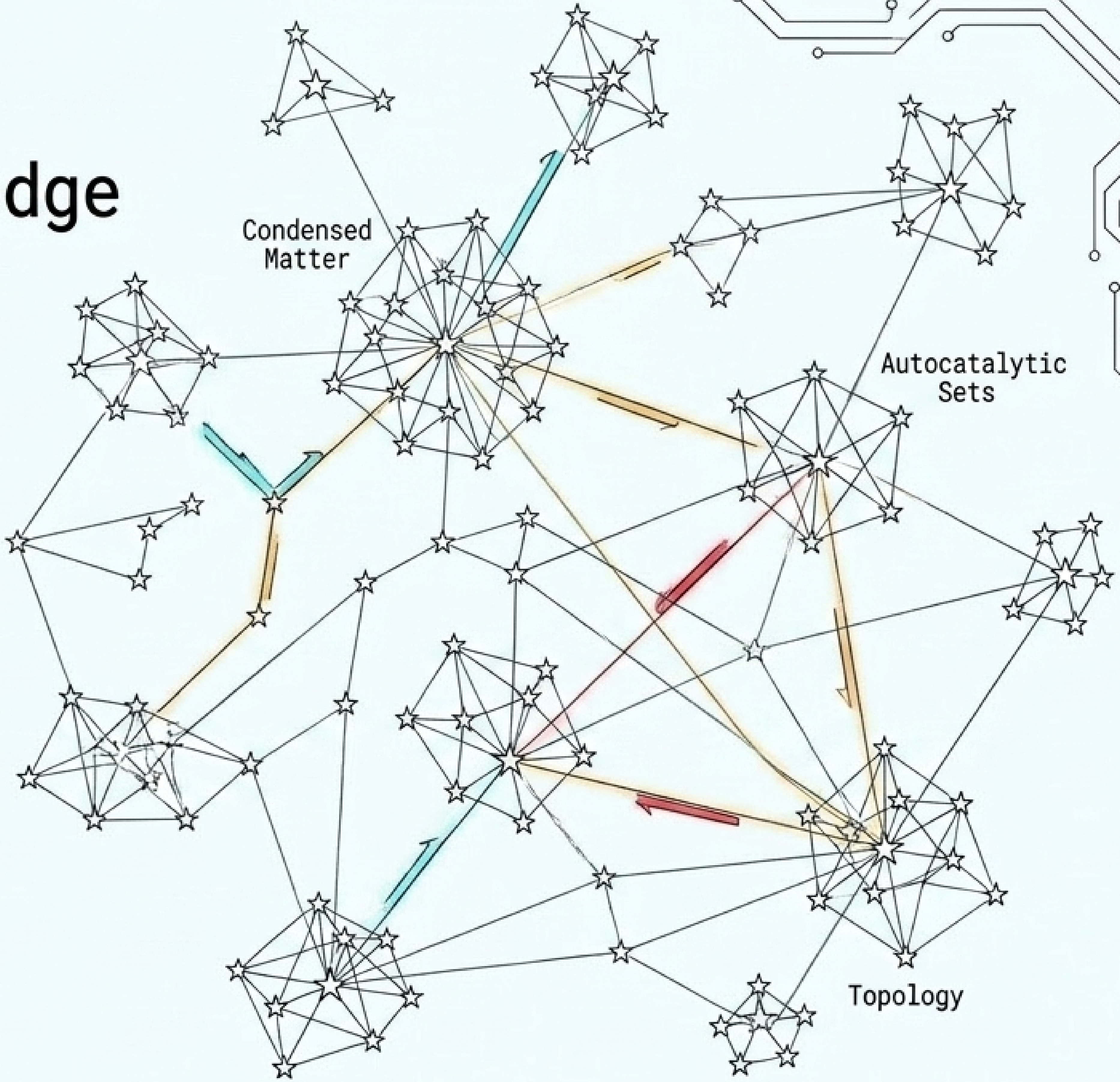
Haplopraxis: A Topology of Knowledge

A massively distributed archaeology game where the map is exactly as large and structured as the source knowledge graph.

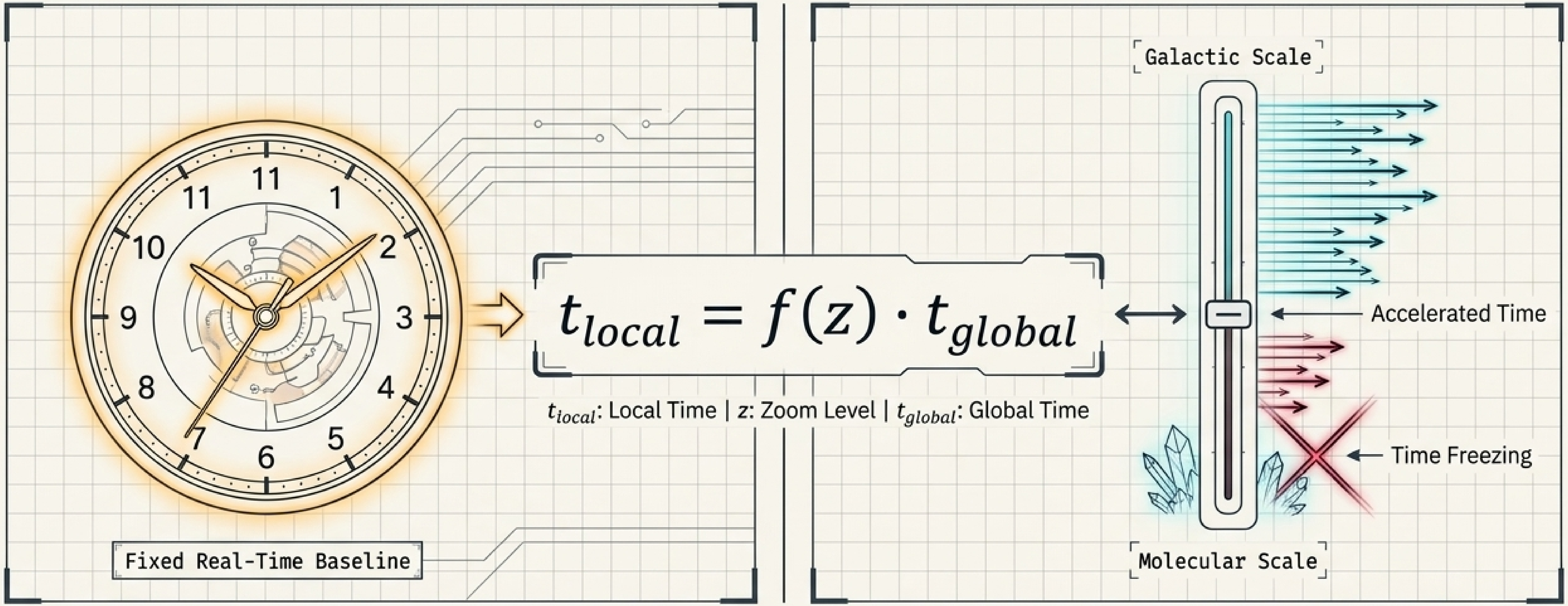
- Knowledge isn't a decoration applied to gameplay; knowledge is the geography.

- No privileged entry point (Rhizomatic exploration).

- Visuals are outline-only and blur without attention: perception must be continually maintained, not passively received.



Time as a Property of Perspective



$$t_{local} = f(z) \cdot t_{global}$$

t_{local}: Local Time | z: Zoom Level | t_{global}: Global Time

Zooming scales time. Time stops being a universal coordinate—two players at different zoom levels inhabit different effective presents. Synchrony is the game's scarcest resource, requiring time crystals to reconcile local history with shared global history.

The Physics of Evaluation

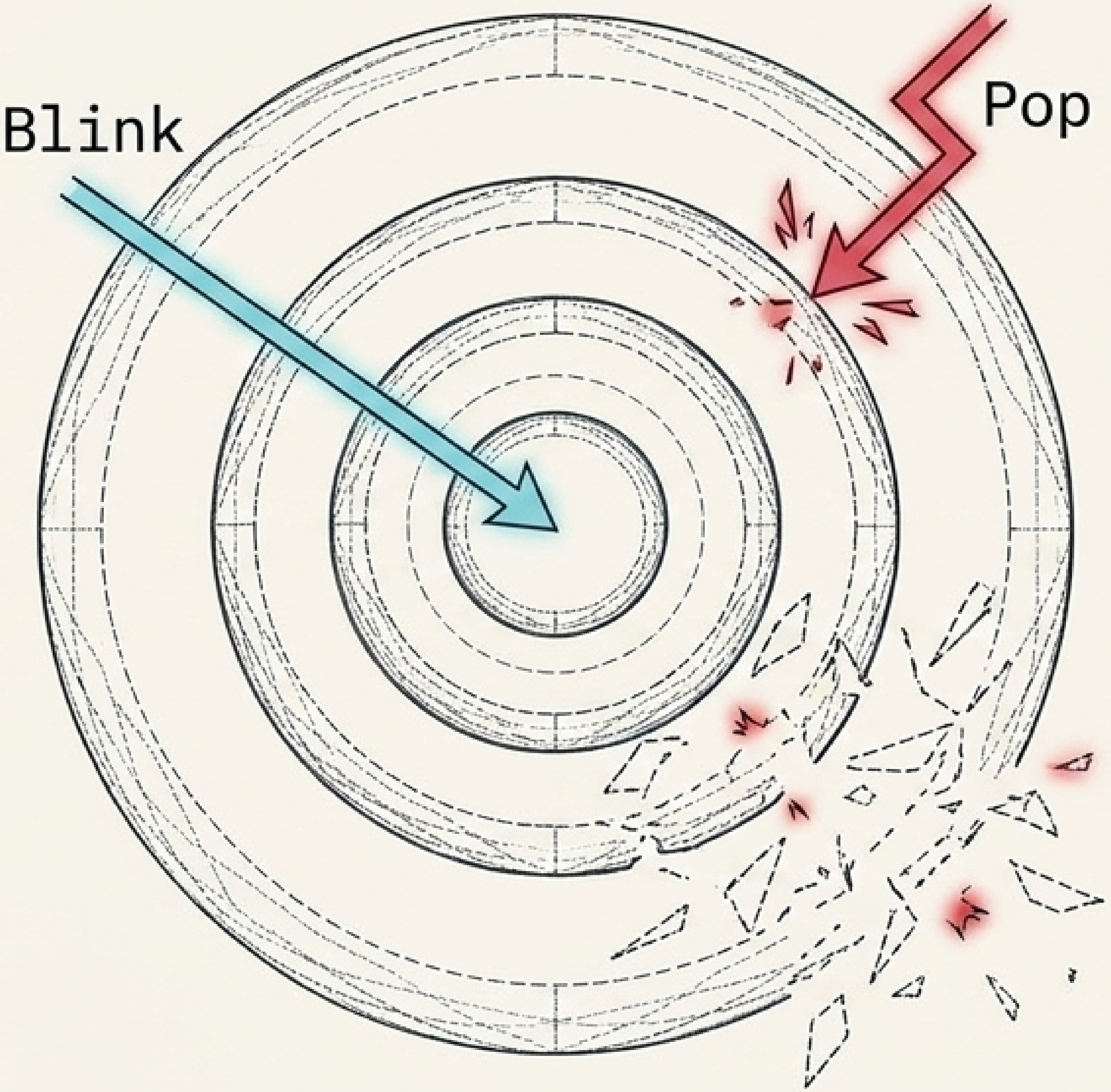
POP

REFUSE

BIND

COLLAPSE

SEEN



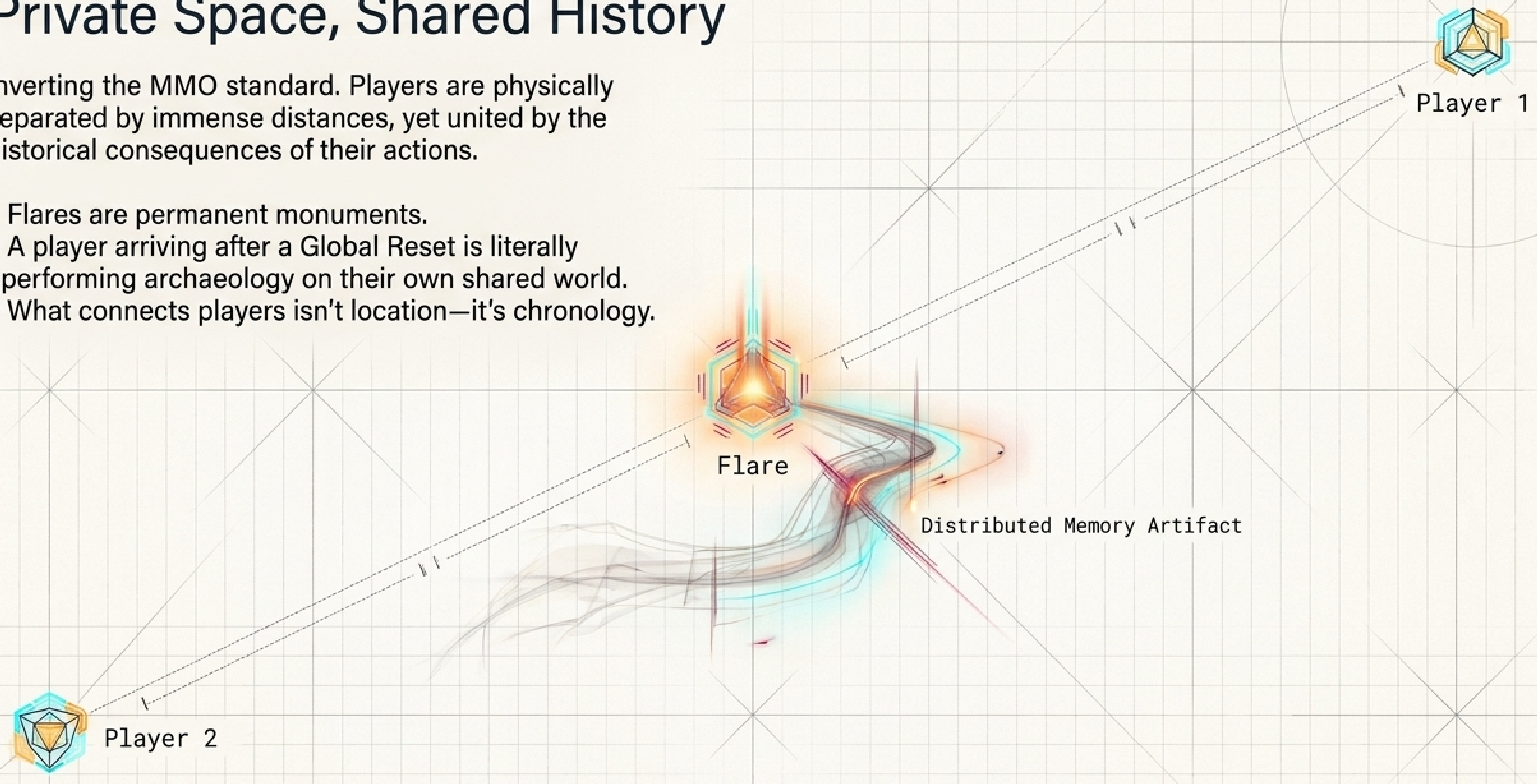
The event vocabulary functions as physical laws. A careless traversal triggers triggers the outer scope and destroys access to the inner scope.

This formal principle (innermost-first evaluation) generates algebra tutorials, stealth mechanics, and diplomacy simultaneously.

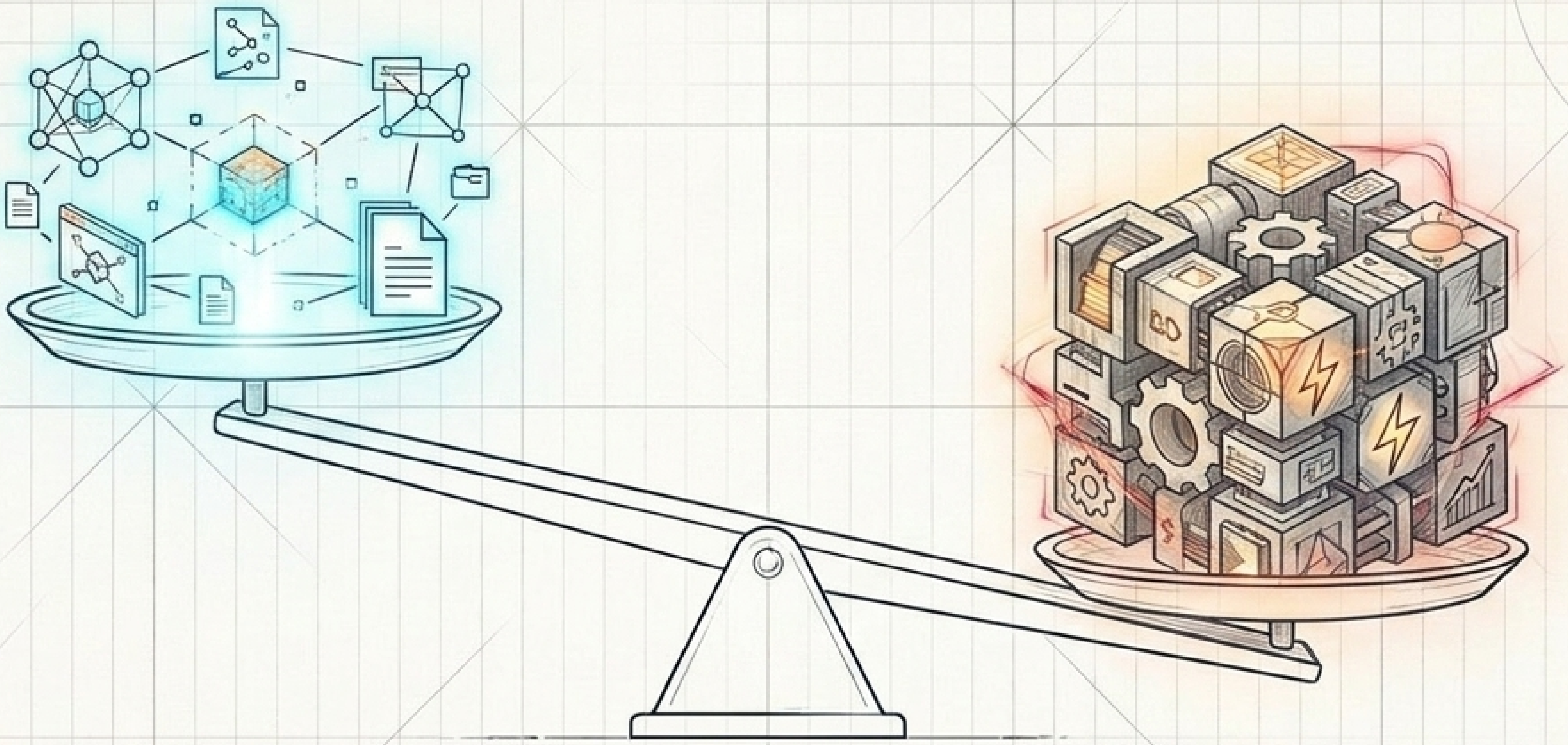
Private Space, Shared History

Inverting the MMO standard. Players are physically separated by immense distances, yet united by the historical consequences of their actions.

– Flares are permanent monuments.
– A player arriving after a Global Reset is literally performing archaeology on their own shared world.
– What connects players isn't location—it's chronology.



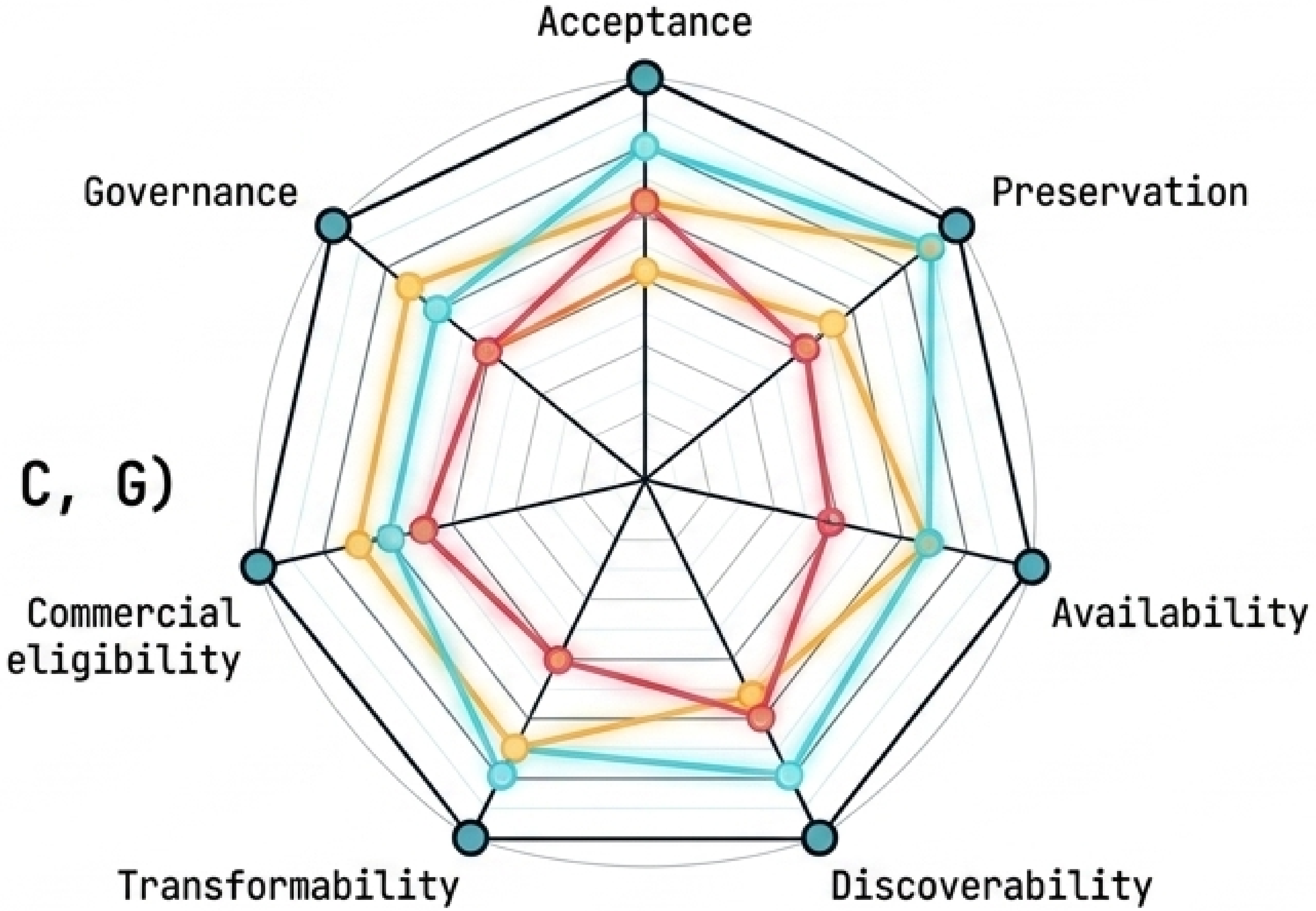
The Principle of Informational Abundance



Information should become easier to preserve and access as collective participation increases. Uploading isn't paid. Compensation attaches only to verified service (storage, indexing, computation, physical fabrication). Revenue from physical realization sustains the infrastructure that keeps the digital commons free.

The Orthogonal State Vector

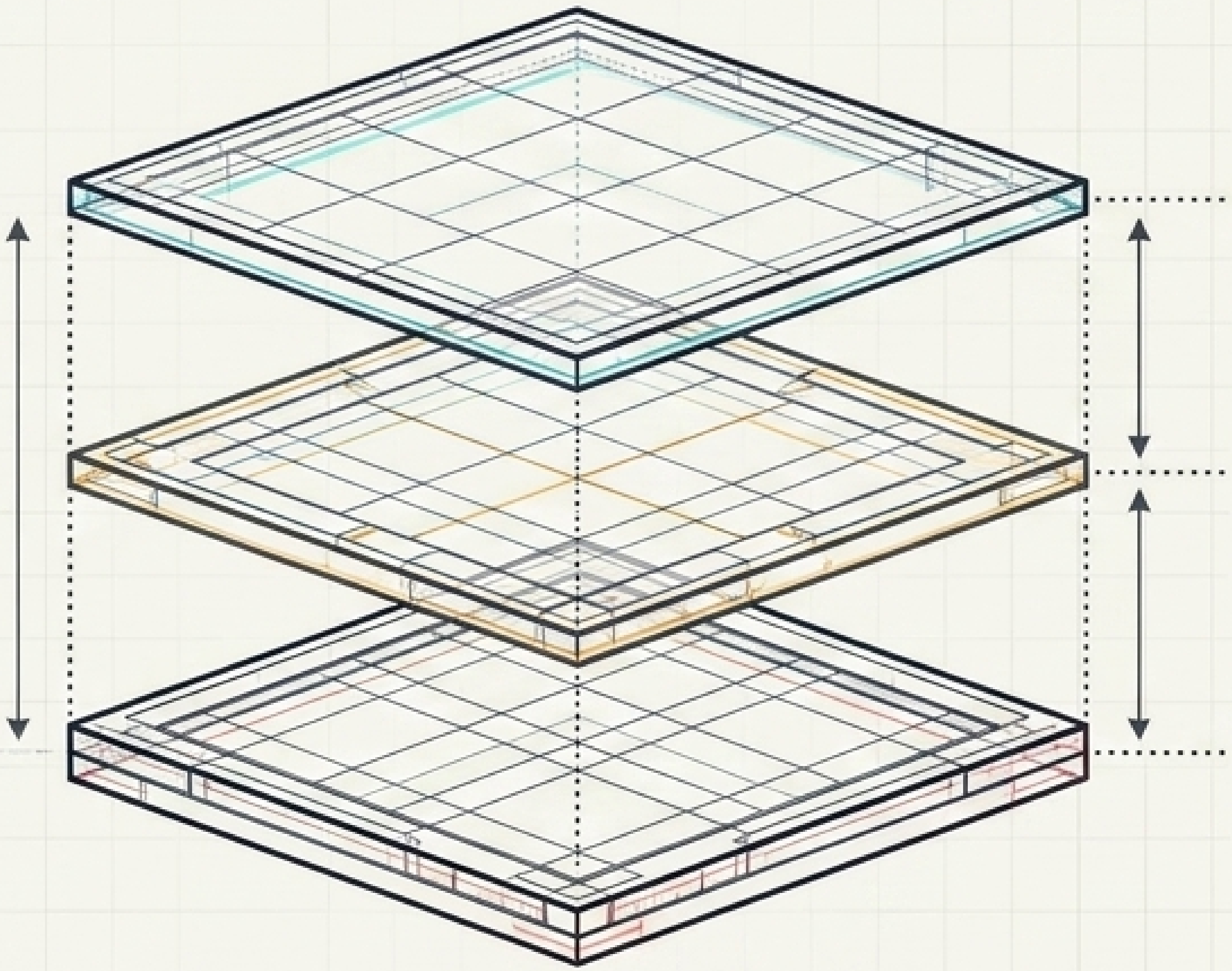
$$O = (A, P, V, D, T, C, G)$$



A:	Acceptance
P:	Preservation
V:	Availability
D:	Discoverability
T:	Transformability
C:	Commercial eligibility
G:	Governance

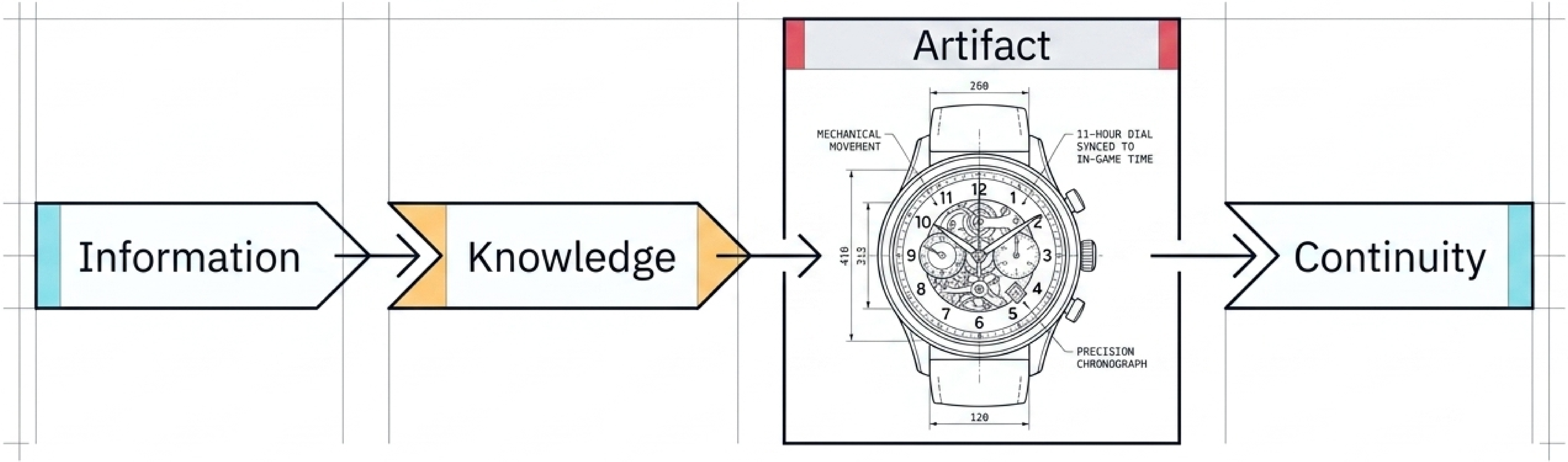
Most systems collapse state into “exists/doesn’t” or “public/private.” URFN treats state orthogonally. A fle can be Preserved and Available, while remaining completely non-Discoverable and non-Commercial.

De-Tangling Identity from View



Semantic location is explicitly not canonical identity; it is a derived view. The archive must be re-indexable without being re-identified. Decoupling semantic placement from the payload ensures archive integrity outlasts any single machine learning model's lifespan.

Continuity Through Realization



Physical goods (e.g., printed books, 11-hour watches synced to in-game time) are not merchandise add-ons. They are the terminal points of the network. A physical artifact is a mechanism of continuity that survives node churn, model obsolescence, and platform collapse.

The Relational Ontology Matrix

	LLM Fine-Tuning	Haplopraxis	URFN Protocol
- "Core Constraint"	Admissibility Bottlenecks	Scarcity of Synchrony	Cost of Abundance
- "Definition of State"	Relational Routing ($z(i) \Rightarrow j$)	Event Append Log (Flares)	Orthogonal Vector (O=A,P,V,D,T,C,G)
- "Canonical Identity"	Position in Graph	Chronology of Traversal	$I_{payload} + I_{history}$ (Not Semantics)
- "Mechanism of Time"	Checkpoint Epochs (T_{gen})	Zoom-Scaled Local Time	Phased Realization

Across AI math, indie game physics, and decentralized protocols, the underlying philosophy holds: State is a derived view; history and relational routing are primary.

Architectures of History

1. State is a summary; history is the world.

The event log is the only true ontology.

2. Knowledge requires a route, not just a space.

Encoding a fact is useless if it is not admissible to the the operator that needs it.

3. Persistence demands maintenance.

Memory, perception, and continuity and only survive through active reconstruction and physical realization.

The system architect's blueprint for relational continuity.